

实验四：物联网监控系统

实验内容

使用websocket建立网络聊天室，聊天室成员既包括web用户，也包括物联网节点，web用户可对物联网节点进行查询和控制。物联网节点可以是虚拟节点，如：mqtt.fx。

准备工作

网络聊天室仿照实验三部署在云服务器。

本地windows系统安装mqtt.fx。

实验步骤

编写服务端代码

```
const WebSocket = require('ws');
const wss = new WebSocket.Server({ port: 8080 });

const clients = new Map(); // 存储每个连接的信息

wss.on('connection', (ws) => {
  ws.on('message', (message) => {
    const data = JSON.parse(message);

    if (data.type === 'register') {
      clients.set(ws, { id: data.id, role: data.role });
      console.log(`[注册] ${data.role}: ${data.id}`);
    }

    if (data.type === 'chat') {
      // 广播聊天消息
      for (let client of clients.keys()) {
        if (client.readyState === WebSocket.OPEN) {
          client.send(JSON.stringify(data));
        }
      }
    }

    if (data.type === 'command') {
      // 用户向节点发送控制命令
      for (let [client, info] of clients.entries()) {
        if (info.id === data.target && info.role === 'node') {
          client.send(JSON.stringify(data));
        }
      }
    }
  });
});
```

```

        break;
    }
}
}

if (data.type === 'status') {
    // 节点返回状态，发送给用户
    for (let [client, info] of clients.entries()) {
        if (info.id === data.target && info.role === 'user') {
            client.send(JSON.stringify(data));
            break;
        }
    }
}
});

ws.on('close', () => {
    const info = clients.get(ws);
    console.log(`[断开] ${info?.role}: ${info?.id}`);
    clients.delete(ws);
});
});

console.log("WebSocket server is running on ws://localhost:8080");

```

编写前端界面

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>聊天室与物联网控制台</title>
</head>
<body>
    <h2>Web用户聊天室</h2>
    <input type="text" id="msgInput" placeholder="输入消息..." />
    <button onclick="sendChat()">发送</button>
    <br><br>

    <input type="text" id="nodeId" placeholder="设备ID" />
    <input type="text" id="cmdInput" placeholder="控制命令" />
    <button onclick="sendCommand()">发送命令</button>

    <h3>聊天/状态输出: </h3>
    <pre id="output"></pre>

    <script src="script.js"></script>

```

```
</body>
</html>
```

```
const ws = new WebSocket('ws://139.155.108.193:8080');

ws.onopen = () => {
  ws.send(JSON.stringify({ type: 'register', id: 'webUser1', role: 'user'
}));
};

ws.onmessage = (event) => {
  const data = JSON.parse(event.data);
  const output = document.getElementById('output');

  if (data.type === 'chat') {
    output.textContent += `[聊天] ${data.from}: ${data.message}\n`;
  } else if (data.type === 'status') {
    output.textContent += `[状态] 来自设备${data.id}: ${data.status}\n`;
  }
};

function sendChat() {
  const msg = document.getElementById('msgInput').value;
  ws.send(JSON.stringify({ type: 'chat', from: 'webUser1', message: msg }));
}

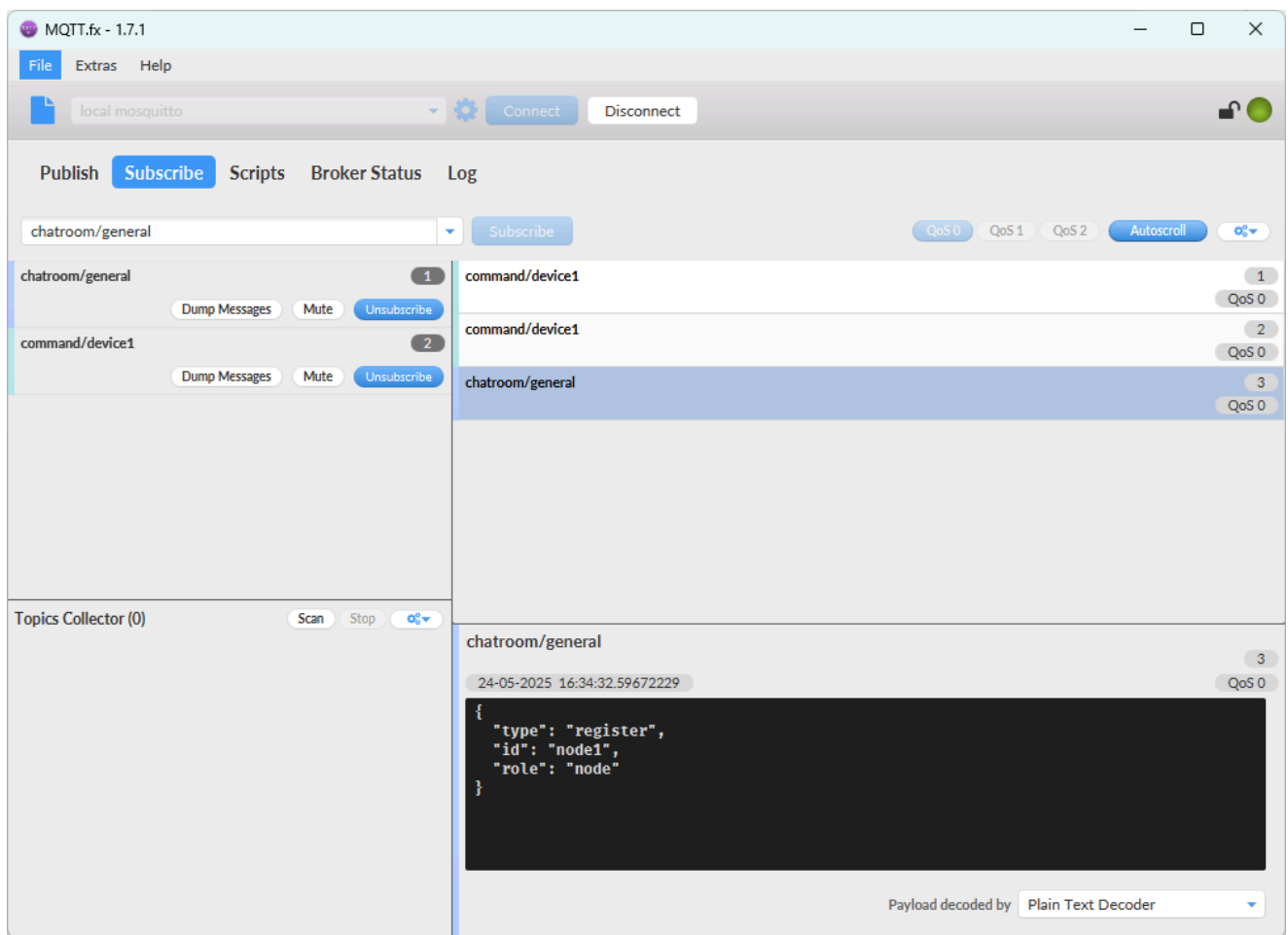
function sendCommand() {
  const target = document.getElementById('nodeId').value;
  const cmd = document.getElementById('cmdInput').value;
  ws.send(JSON.stringify({ type: 'command', id: 'webUser1', target: target,
command: cmd }));
}
```

本地安装 mqtt.fx

新版本需要付费使用，安装旧版本即可。

<https://iotcore-dev-tool.gz.bcebos.com/mqttx/1.7.1/mqttx-1.7.1-windows-x64.exe>

用 mqtt.fx 订阅、发布



编写桥接代码

```
const WebSocket = require('ws');
const mqtt = require('mqtt');

const wss = new WebSocket.Server({ port: 8081 }, () => {
  console.log('WebSocket 服务器已启动: ws://139.155.108.193:8081');
});

const mqttClient = mqtt.connect('mqtt://localhost:1883');

wss.on('connection', (ws) => {
  console.log('网页客户端已连接');

  // 接收来自网页的消息
  ws.on('message', (msg) => {
    try {
      const data = JSON.parse(msg);
      if (data.type === 'chat') {
        mqttClient.publish('chatroom/general', msg);
      } else if (data.type === 'command') {
        mqttClient.publish(`command/${data.target}`, msg);
      } else {
        console.log('未知消息类型', data);
      }
    }
  });
});
```

```
    } catch (e) {
      console.error('消息解析错误:', e);
    }
  });

  // 当从 MQTT 收到消息后，发送给 WebSocket 客户端
  mqttClient.on('message', (topic, message) => {
    ws.send(message.toString());
  });

  // MQTT 连接成功后订阅主题
  mqttClient.on('connect', () => {
    console.log('已连接到 MQTT Broker');
    mqttClient.subscribe('status/+'); // 所有设备状态
    mqttClient.subscribe('chatroom/general'); // 聊天室消息
  });
```

运行示例

Web用户聊天室

123asdsad

发送

设备ID

控制命令

发送命令

聊天/状态输出：

[聊天] webUser1: 123
[聊天] webUser1: 123asdsad

MQTT.fx - 1.7.1

FileExtrasHelp

local mosquittoConnectDisconnect

PublishSubscribeScriptsBroker StatusLog

command/device1SubscribeQoS 0QoS 1QoS 2Autoscroll

chatroom/general1Dump MessagesMuteUnsubscribe

command/device13Dump MessagesMuteUnsubscribe

Topics Collector (0)ScanStop

command/device11QoS 0

command/device12QoS 0

chatroom/general3QoS 0

command/device14QoS 0

command/device14

24-05-2025 16:49:17.60557127QoS 0

ok

Payload decoded byPlain Text Decoder